

The SeaBED AUV – A Platform for High Resolution Imaging

Hanumant Singh, Ryan Eustice, Chris Roman, Oscar Pizarro,

Deep Submergence Laboratory, MS # 7,
Woods Hole Oceanographic Institution,
Woods Hole MA 02543

Abstract

The SeaBED autonomous underwater vehicle (AUV) is a new imaging platform designed for high resolution optical and acoustic sensing. This low cost vehicle has been specifically designed for use in waters upto 2000 meters to carry out video transects, bathymetric and side-scan sonar surveys. In this paper we detail the systems issues associated with navigation, control, and imaging that led us to our particular hardware and software design choices so as to allow us to operate in shallow, shelf and ocean basin environments.

I. Introduction

Affordable access to the seafloor for high-resolution imaging remains an elusive goal for researchers and users in the marine archaeology, marine biology, physical oceanography, marine geology and geophysics and naval communities. These disciplines all share a common need for high quality optical and acoustic surveys from small vessels of opportunity in depths extending just beyond diver attainable depths to those associated with deep ocean basins. The technologies associated with manned submersibles, remotely operated vehicles (ROVs) and towed vehicles that service such needs today, are expensive, require large ships and infrastructure, and are in short supply. The technologies associated with AUVs are rapidly evolving to fill the requirements of these communities. A large number of AUVs are being designed, built and deployed in the support of such tasks. In the US there are a number of ongoing efforts, including those associated with the MIT Odyssey class AUVs (and its commercial successors manufactured by Bluefin Robotics)[1] for oil and gas surveys and naval applications, the ABE AUV[2] designed for deep ocean scientific surveys, the Altex AUV[3] designed for under-ice surveys in the Arctic, and the Ocean Explorer and Morpheus AUVs[4] designed for coastal surveys. Other notable efforts abroad include the British Autosub AUV[5], the Norwegian Hugin AUV[6] and the Japanese Urashima AUV[7].

This paper outlines the design choices for the SeaBED AUV and lays out the results of our first set of test cruises aimed at characterizing deep water coral habitats. Section II goes over the vehicle components – mechanical, sensors, power, and computing. Section III is dedicated to descriptions of the navigation and control systems with Section IV offering concluding remarks.

II. Basic Vehicle Components

II.1 Mechanical

The Seabed vehicle (Figure 1) is composed of two torpedo like body sections fixed to each other with vertical structural members. Each of the hull section are 1.9 meters long and .34 meters in diameter. The separation between the hulls, centerline to centerline, is 1.1 meters. The width of the vehicle, measured to the outside of the main thrusters, is .9 meters. The overall weight of the vehicle is 200 kg. The internal parts of the vehicle are covered with ABS plastic skins to create the two torpedo shaped hulls.



Figure 1. The SeaBED AUV

The top hull of the vehicle contains the foam flotation and the main electronics housing which is also positively buoyant. The three main pressure housings on the vehicle - the electronics housing, the battery housing and the side scan sonar computer housing, are all constructed from 7075-T6 Al and have hemispherical end caps. Hemispherical end caps were chosen, in preference to flat endcaps, for the significant weight savings associated with such a design for housings over 15 cm in diameter. The foam flotation used is molded to shape and has a specific gravity of 0.4. The electronics in the main housing are connected to the other components by wet cabling routed through the vertical struts. The vertical struts have an airfoil profile and are oriented with the preferred direction of travel forward. Drag tests performed with a scale model indicated that the drag produced by the vertical airfoil shape was significantly less than the cylindrical shape originally envisioned. The main thrusters are also mounted with brackets made from airfoil shaped struts.

II.2 Sensors and other Systems

The bottom hull of the vehicle contains all of the sensors with the exception of the Crossbow IMU that is mounted inside the main electronics housing and consists of angular gyroscopic rate sensors, accelerometers and a fluxgate compass. Most of the components in the bottom hull are negatively buoyant. The following list of sensors are currently integrated into the vehicle design.

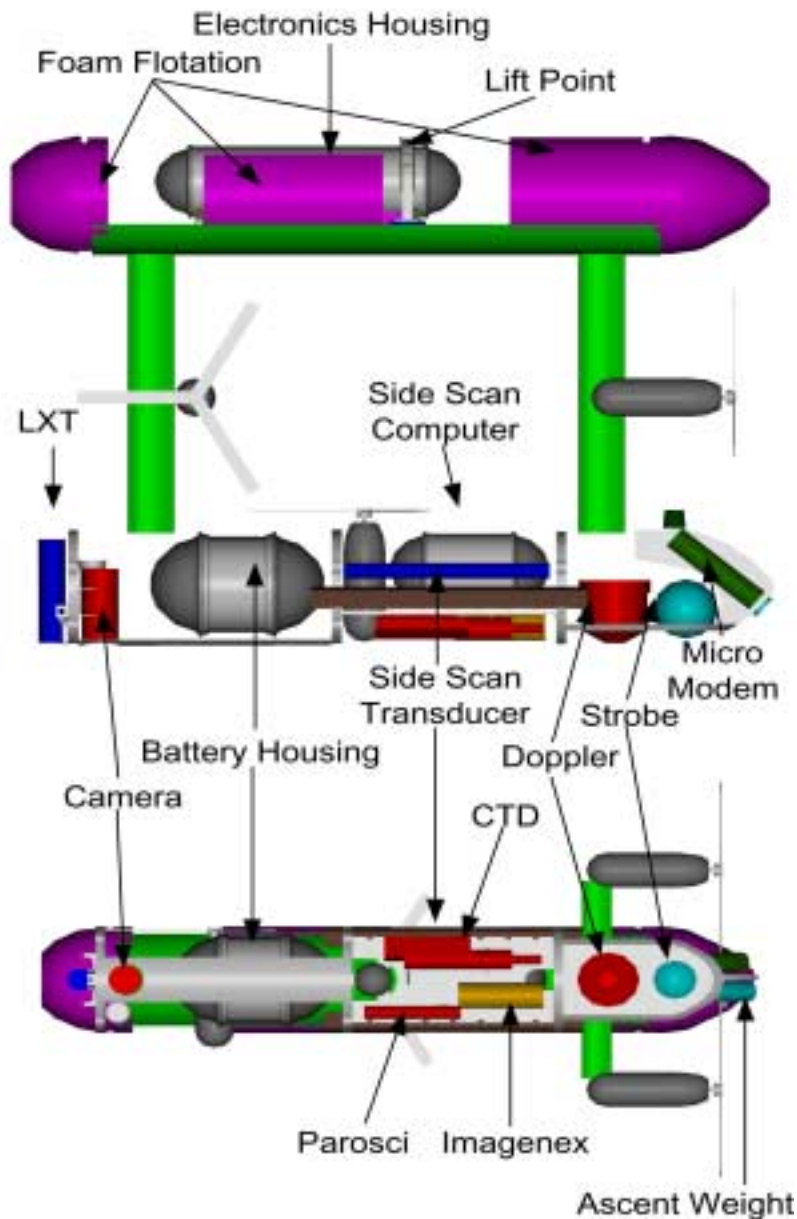


Figure 2. Vehicle cad model showing the layout of the various sensors as well as the main mechanical components.

- A high dynamic range digital still camera is the primary optical imaging sensor. This is a Pixelfly 1024x1280 12 bit CCD camera mounted in a flat glass plate housing. An example image using this camera on Seabed off of Puerto Rico in a deep sea coral habitat is shown in Figure 3.

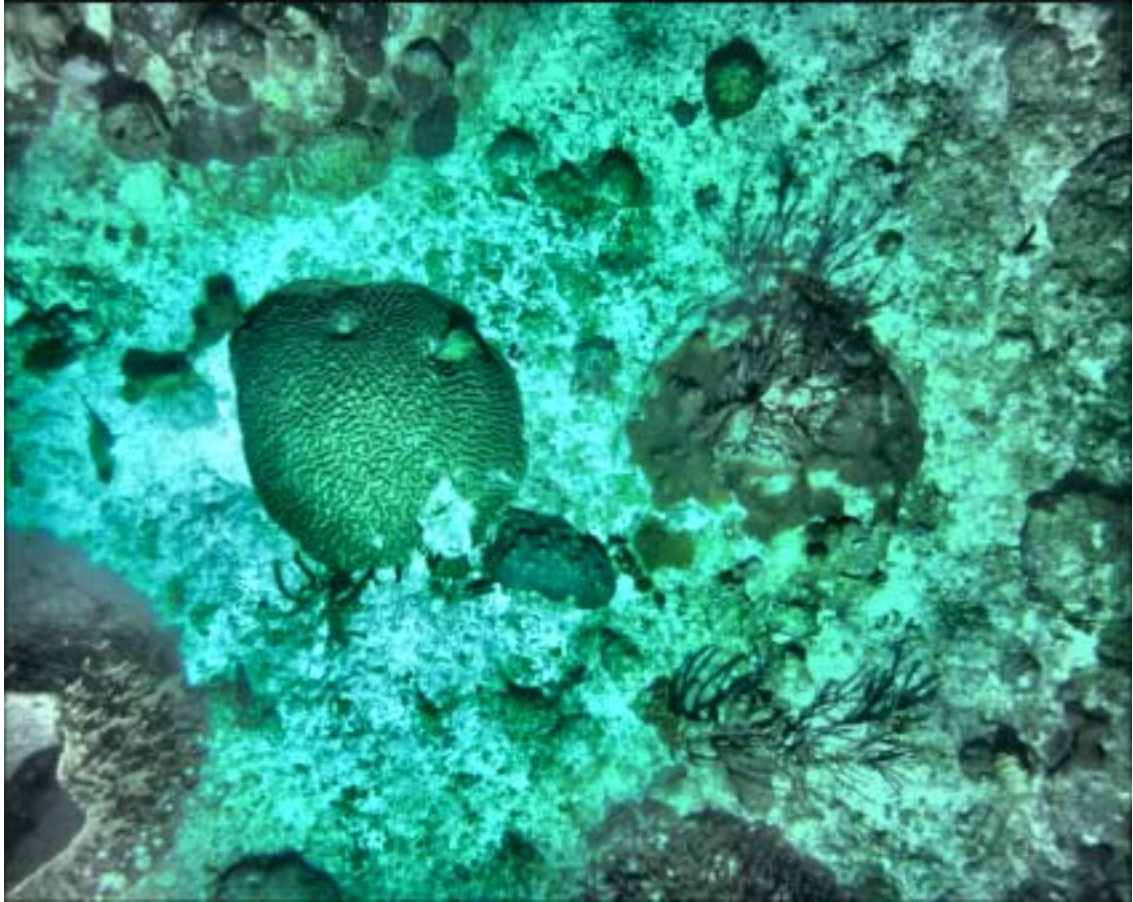


Figure 3. This image showing different species of coral along the shelf edge off of Puerto Rico was taken with the SeaBED AUV.

- A 300 kHz side scan sonar system (from Marine Sonics Technologies) with 6000 meter rated transducers . This system operates on its own PC-104 stack running the Windows operating system. This system is controlled in its operation from the main vehicle computer system. Example data from this system in shallow water is shown in Figure 4.
- An RDI Workhorse Navigator 300 kHz ADCP is the primary navigation sensor and is operated in bottom lock mode. The ADCP provides measurements of velocity over the bottom, heading, altitude, pitch, roll, and integrated position. The doppler position estimate is accurate to 1-5% of the distance traveled.
- A 150 watt-second strobe is used for photographic illumination. The strobe is mounted 1.4 meters behind the camera to reduce the effects of lighting backscatter in the images.
- An Imagenex 881 scanning head pencil beam sonar is used to collect bathymetric data. The sonar frequency is variable between 300 kHz and 1.2 MHz. The sonar is mounted to scan athwart ships and capable of scanning ranges upto 200 meters

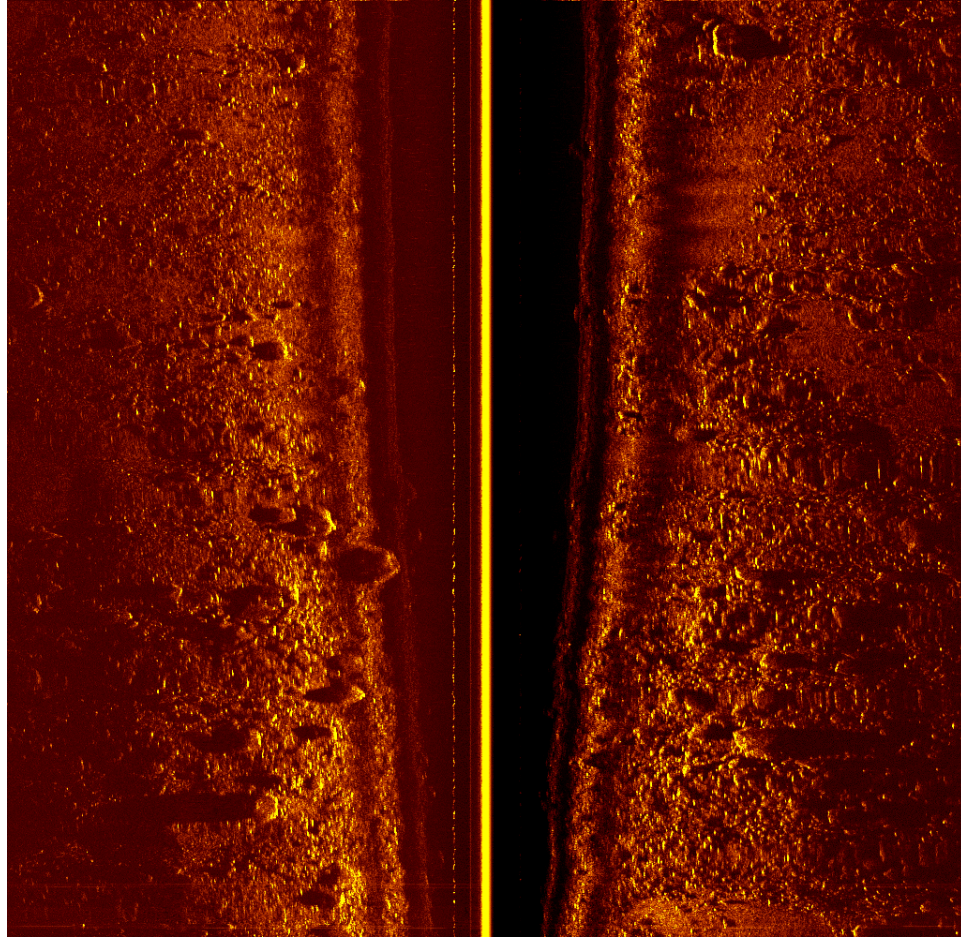


Figure 4. Side scan sonar data obtained with the SeaBED AUV. The site is off of Woods Hole, MA and was used as a bombing range towards the end of the Second World War.

- A Paroscientific depth sensor provides depth information with an accuracy of centimeters.
- A Seabird CTD sensor
- An underwater acoustic modem is integrated in the vehicle to provide LBL support and low bandwidth communication with the surface. At present, the modem is programmed to send 32 byte data packets back from the vehicle and listen for abort commands sent from the surface to the vehicle while it is operating.
- An Ascent/Descent weight system is used to help the vehicle transit to the bottom and provide a factor of safety for the vehicle ballasting. Typically the vehicle is ballasted one pound buoyant to allow the vehicle to float up to the surface in case of a system emergency.
- Tracking and Emergency. The vehicle carries an acoustic ultra-short baseline tracking beacon (from ORE corporation) to provide range and bearing at a top side display. This information is used only to provide an approximate location of the vehicle during a mission. An emergency strobe and emergency RF tracking beacon are used at all times to help aid in the recovery of the vehicle.

II.3 Thrusters

The four thrusters used to propel the vehicle are identical DC brushed motors enclosed in one atmosphere pressure housings. The housings use a magnetic coupling between the motor and the shaft for the propeller, avoiding the need for high pressure shaft seals which typically have high levels of friction. The motors turn 24" diameter graphite propellers with 12 inches of pitch.

The thruster motors are torque controlled using a PWM switching controller. Operating in open loop, a specified motor current will result in a corresponding vehicle velocity through the water in a steady state situation. In closed loop control, the desired motor current is manipulated to achieve a desired vehicle speed over ground. Vehicle speed is measured with the Doppler ADCP. A simple relationship between thruster motor current and the actual thrust produced was determined using a Bollard type test with a calibrated load cell. Using these motors and propellers we have been able to control the vehicle to do survey work at speeds between zero and one meter/second. In general, accurately controlling slower vehicle speed is more difficult given the complex relation between thruster motor current and the actual thrust produced around the zero operating point.

II.4 Power

The vehicle carries a compact 2 kWh battery. The pack consists of 126 high-energy density Li-ion secondary D cells (from Eagle Pitcher Corporation) and the necessary circuitry for monitoring and recharging these cells without venting the battery housing. The battery pack is organized in 14 serial stacks. Each individual stack is made up of nine cells connected in parallel. The monitoring circuitry provides real-time voltage and charge capacity measurements for our battery pack.

The main vehicle power varies between 56V and 42V and is fed to a number of high input range DC-DC converters that convert this voltage to the appropriate bus voltages (5V, 12V, 24V, 48V) required for the computing resources, sensors and thrusters. We have taken pains to provide separate isolated power supplies to prevent electrical noise from corrupting different sensors. While this may be considered wasteful, as the individual converters have power losses associated with efficiency (even considering in our case that we turn them off with relays if they are not required) the payoff lies in our ability to seamlessly integrate a large variety of sensors. For example, the sidescan sonar and the ADCP both operate at 300kHz. However they do not interfere with each other and the fact that we can run them asynchronously is a testament to the isolation that we have achieved through our power bus design.

II.5 Computing

The main computing source on the vehicle is a PC-104 180 MHz Pentium system running Redhat Linux 6-2. The Linux operating system has proved a stable basis for the operation of the AUV as well as been a boon for software development.

The vehicle control loop runs at 10 Hz. The basic software architecture follows a multi threaded approach, with individual threads designated for hotel management, control execution, sensor support, data logging and mission command. Operation of the vehicle consists of a main program to take care of sensors, hotel, logging and control running in parallel with a mission planning program which governs the higher level task execution specific to a given mission. Missions are defined in a mission script format consisting of a sequence of behavior specific tasks and related hotel information. The mission planning program (written in Perl) communicates with the main program through a socket connection by sending formatted messages. This split between a mission planner and main routine has helped separate the higher and lower level tasks of vehicle operation. The messages sent by the mission planner are composed of lower level goals that keep the main software separated from detailed task specific support.

Vehicle data is logged during operation to ASCII formatted files that are organized by date and mission name. The core navigation data is logged at 5 Hz. The camera images generate the most significant amount of data. At the maximum rate of an image every 2.5 seconds, determined by the speed images can be written to the hard drive and the charging time required by the strobe, the camera generates ~1Mbyte/sec. The sidescan sonar data is stored on the PC-104 computer dedicated to the sidescan system.

III. Navigation and dynamics

III.1 Vehicle Shape

The SeaBED vehicle is shaped to create a passively stable platform in roll and pitch. The separation of the two torpedo shaped hulls, the top containing syntactic foam flotation and the bottom containing heavy components, has many advantages for an imaging platform. The vehicle is very stiff in roll and pitch while still easily controlled in the four other degrees of freedom. This separation also minimizes the effects of body lift and body roll which are associated with single hull torpedo style AUVs. This configuration also helps decouple the other controllable axes, making control design and development a simpler task. The theoretical estimate for the metacentric height, the separation between the CG (center of gravity) and CB (center of buoyancy) is approximately 23 cm. This value for metacentric height is significantly more than the typical 2-5 cm separations associated with most torpedo shaped vehicles.

Typical pitch and roll excursion for the the vehicle, when excited by its own thrusters, are less than two degrees and often less that one degree in steady state situations. This behavior is illustrated in Figure 5, which shows approximately 10 minutes of pitch and roll data during a typical survey over a relatively flat bottom. The roll motion is slightly excited as the vehicle descends to depth. The pitch and roll natural frequencies for the vehicle are both approximately 0.3 Hz.

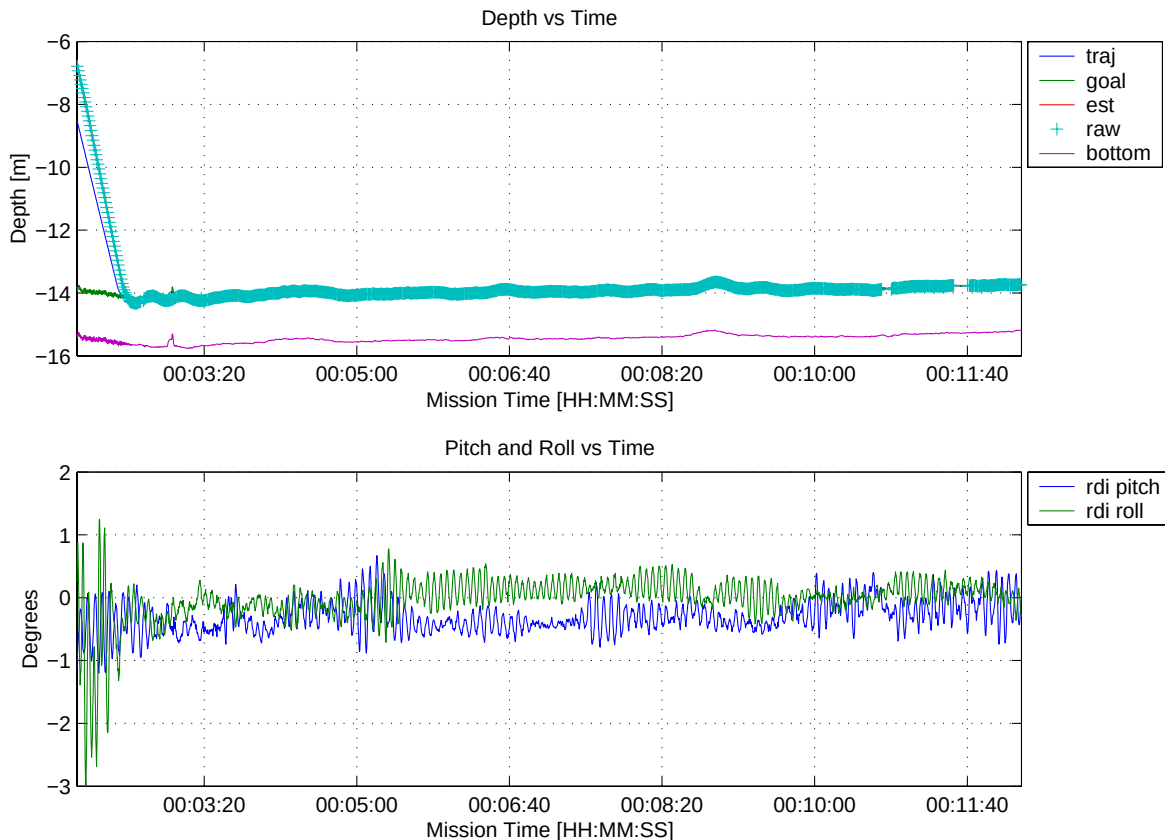


Figure 5. Pitch and roll for a typical mission profile. The large metacentric height of the vehicle makes SeaBED a very stable platform.

III.2 Control System

The basic control architecture is set up similarly for each controlled axis, heading, speed and depth. These axes are then controlled independently with individual servos. In each case the servos are setup to handle goal and reference states. The goal states are manipulated at the level of a desired vehicle behavior. An example would be flying the vehicle at a desired (goal) altitude. The goal states are then filtered with a simple model of the vehicle dynamic response to create a reference trajectory that is achievable by the vehicle. The reference state is then fed to the axis servo controller.

Shaping the reference trajectory with an approximate vehicle response is a simple way of avoiding actuator saturation, typical of commanding references that the vehicle cannot achieve. Also, this decoupling of goal and reference make the transition between different modes of controller operation seamless, such as the transition between flying at a fixed depth and flying to maintain a specified altitude above the bottom.

The large majority of control computation involves the creation of servo references from desired goals. The reference trajectory generation is decoupled into a depth servo and heading and speed servos. The depth trajectory uses the desired goal, depth measurement and altitude measurement to generate a depth reference. The heading and speed trajectories are more complicated and maybe coupled. The vehicle behaviors in the XY plane consist of three basic types; goto a point (x,y), goto a point (x,y) along a line and goto a point (x,y) along a line while maintaining a specific heading. In the first case the heading and speed trajectories are decoupled and the vehicle motion is subject to water current disturbances. In a typical situation the vehicle will trace out a curved "fishhook" path as it approaches the desired end point. The second situation requires the trajectory generation to account for the presence of a current disturbance. In this case we use a vector field, which surrounds the desired trackline, to generate the desired vehicle velocity that will keep the vehicle traveling along the line. Figure 6 shows an example of this vector field for a 20 meter east west trackline.

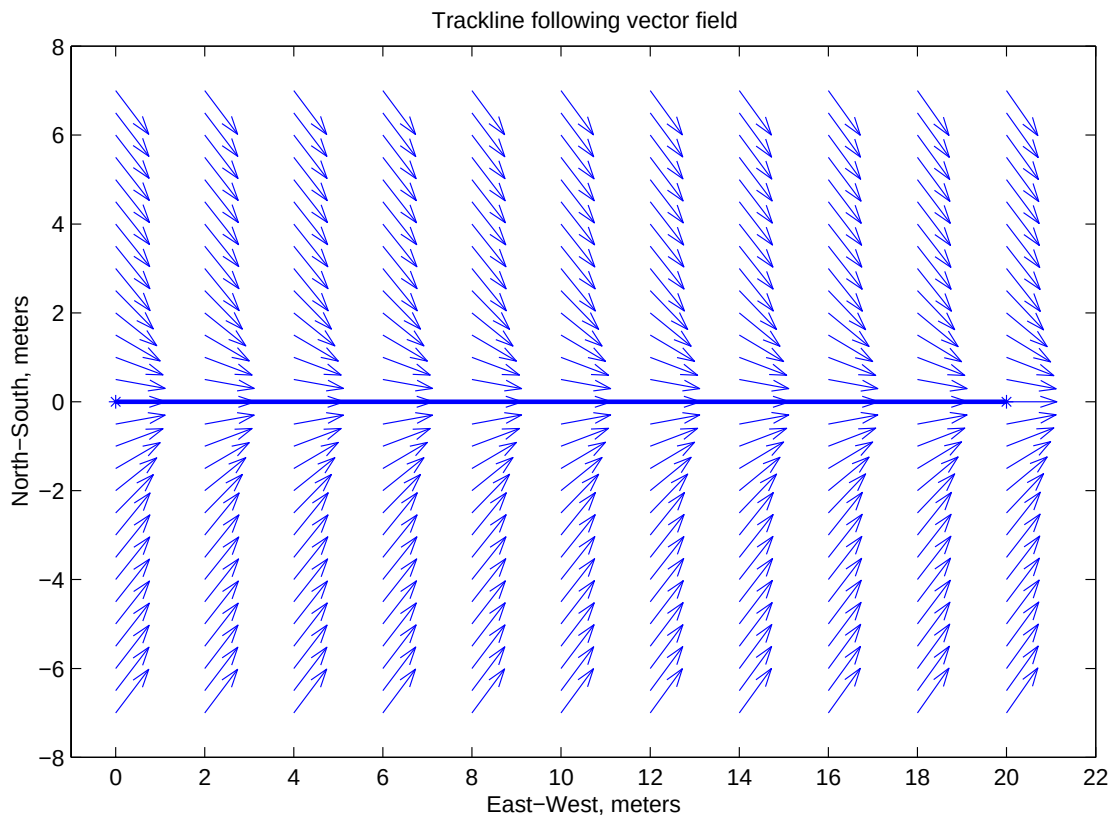


Figure 6. The vector field plot to generate a trajectory along a desired trackline

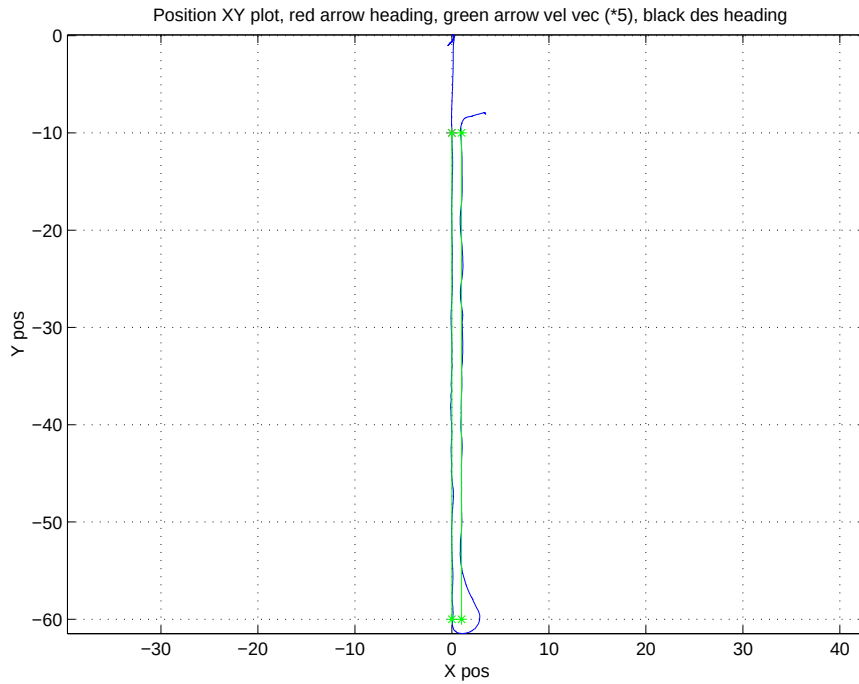


Figure 7. The XY plot for a survey mission consisting of two parallel tracklines spaced one meter apart. The vehicle can achieve such commanded trajectories accurate to a few cm.

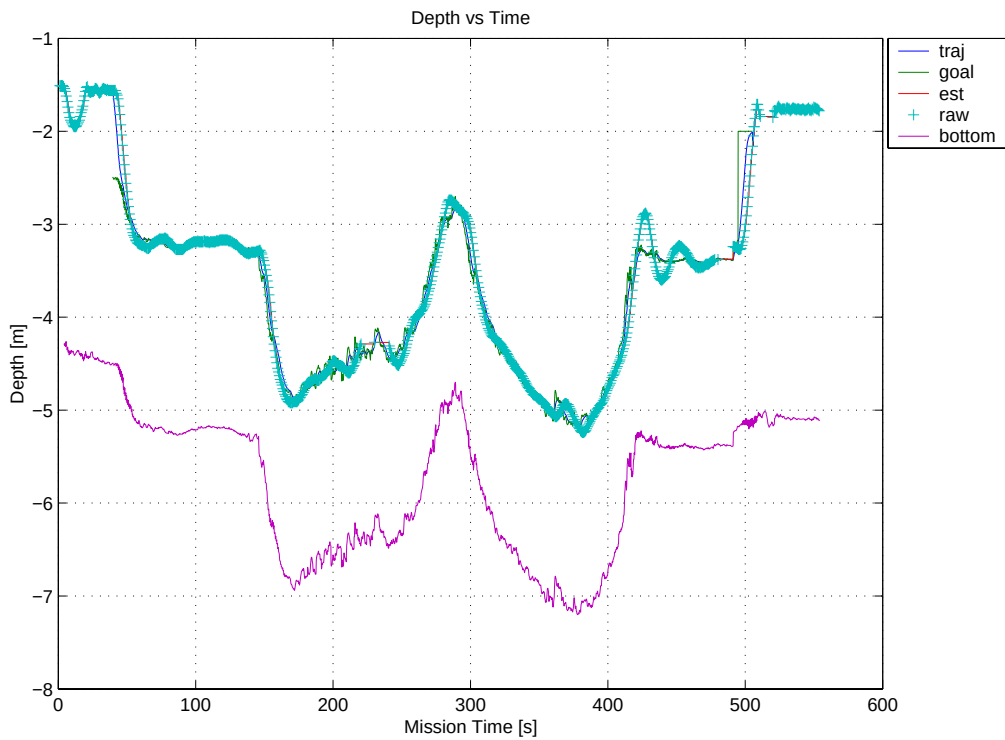


Figure 8. The depth plot for a typical survey mission shows that the vehicle can easily follow the bottom at a fixed altitude within a few cm.

With the desired vehicle velocity defined by the vector field the heading and speed trajectories are defined to produce this desired velocity. The heading trajectory uses the current heading and velocity to produce a reference heading which accounts for a current disturbance.

Control of each servo is done with decoupled PID controllers with additional feedforward terms. In each case the servo controllers generate a desired thrust which is then sent to individual thruster control loops for each of the four thrusters. The thruster control loop manipulates the motor voltage to achieve a desired thrust, which can be related to the motor current. The thruster control loops run at 100 Hz and are tuned to achieve a desired motor current in approximately 0.1 seconds.

Results from an example survey pattern are shown in Figures 7 and 8. The vehicle was commanded to execute two parallel tracklines spaced 1m apart while maintaining a constant depth from the bottom. The plots show that the vehicle can easily maintain depth and xy position of the order of ~10cm.

IV Conclusions

In this paper we have laid out some of the constraints and resulting design choices for building an AUV for conducting high resolution optical and acoustic survey of the ocean bottom. Over the last two years we have used this vehicle to conduct numerous tests associated with sonar mapping and digital camera surveys. These tests have been conducted off of very small vessels as well as off of ocean going research ships. We are funded and expect to use this AUV for numerous high resolution mapping exercises associated with coral reef habitat characterization, benthic mapping for marine biologists, marine archaeology and marine geology and geophysics.

Acknowledgements

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